



Initiation and Implementation of Honey Bee Breeding Programs

Aleksandar Uzunov, Evert W. Brascamp, Manuel Du & Ralph Büchler

To cite this article: Aleksandar Uzunov, Evert W. Brascamp, Manuel Du & Ralph Büchler (2022) Initiation and Implementation of Honey Bee Breeding Programs, Bee World, 99:2, 50-55, DOI: [10.1080/0005772X.2022.2031545](https://doi.org/10.1080/0005772X.2022.2031545)

To link to this article: <https://doi.org/10.1080/0005772X.2022.2031545>



Published online: 15 Feb 2022.



Submit your article to this journal [↗](#)



Article views: 217



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 2 View citing articles [↗](#)

Initiation and Implementation of Honey Bee Breeding Programs

Aleksandar Uzunov , Evert W. Brascamp , Manuel Du  and Ralph B  chler 

As usual in science, where one answer raises new questions, following our attempt to demystify the basic concept of honey bee breeding with our article *The Basic Concept of Honey Bee Breeding Programs* (Uzunov et al., 2017) questions were raised on details about the implementation of a breeding program. Aspects, such as prioritisation of the traits to select for, selection using breeding values, time management of the breeding program, and mating control, prevailed as most intriguing and are indeed aspects to which previously only limited attention has been given. These particular and relevant questions indicate, however, that the message from our previous article reached our target group. This time, we will address in more detail trait prioritisation, the timing of a breeding program and the selection process. Mating control will only be addressed in a general sense.

Trait Prioritisation

"Which traits should I choose to improve?" is the most important question, as it determines the direction in which traits will change over generations by selection. It is the first step towards the initiation of a breeding program, together with the recording of these traits to allow selection.

Based upon experience and subjective sense, beekeepers have been selecting bees for a very long time, though perhaps not fully systematically. However, with the ground-breaking invention of movable frames over a century ago, that allowed direct access to the queen and her easy replacement, selection of honey bees advanced. It did not only allow easy exchange of the queen, which also meant a change of the genetic composition of the colony, but also enabled a better scientific understanding of the enigmatic biology, and in particular the reproductive biology and genetics, of honey bees. With such tools at disposal, beekeepers started

replacing the queens from the colonies with non-desirable expression of traits with those reared from more productive and gentle colonies. That, to significant extent, reshaped the European diversity by massive propagation and use of queens from the commercially best-known subspecies, *Apis mellifera ligustica* and *A. m. carnica*, and enhanced the importance of morphological traits as a means of origin discrimination. Around half a century ago, methods for evaluation of colony performance were significantly improved, which enabled quantification of expression of the traits of interest (Ruttner, 1972) and introduction and use of advanced statistical models for assessment of the colonies' merit for breeding (Bienefeld et al., 1989). By time, the list of traits became lengthy, as well as the beekeepers' breeding ambitious (B  chler et al., 2013).

The prioritisation of traits is affected by the breeders' interests, which may be very diverse, including population conservation or specific traits improvement, hobby or commercial beekeeping, etc. Until recently, most beekeepers and breeders chose honey production, swarming tendency, gentleness, and overall vitality as prime traits of interest. Currently, however, beekeepers diversify as, on the one hand, they commercialize, but on the other hand, show increased interest in the conservation of native and local populations, and are confronted with new pests and pathogens. Diversifying breeding goals may be the consequence of this. Over time, when some breeding-goal traits, like for instance gentleness, reach a satisfactory level, the priority as to which traits to emphasise may change.

Morphological traits, such as pigmentation, belong to the oldest selection criteria for honey bees. Although they mostly have little or no impact on a colony's performance, there has always been a market demand for bees with certain appearances (Armbruster, 1919). However, also

in breeding programs that focus purely on the economic improvement of the stock, morphological features can play a significant role. This is because honey bee breeding usually takes place within a specified subspecies or breed, and by checking the morphology, one can increase the likelihood that a selection candidate belongs to the intended biological group. The importance that should be attached to these traits depends on the situation. If breeding takes place in a fairly open breed, like Buckfast, a queen's outward appearance may be seen as merely a secondary add-on. An opposite extreme is *A. m. ruttneri*, the dark endemic honey bee of Malta. This subspecies is heavily endangered by hybridisation and the breeding group *Breeds of Origin Conservancy* (www.facebook.com/BreedsOfOrigin) is working on its conservation. Here, alongside the use of novel genetic markers for population determination, such as SNPs (Momeni et al., 2021), the morphological characteristics of the Maltese honey bee form a strict entry criterion for the breeding program.

An additional and recently more proclaimed aspect of honey bee breeding is a local adaptation (Meixner et al., 2015; Kova  i   et al., 2020), referring to well-adapted native or naturalised local populations, which are antithetical to the massive commercialised and widely traded stocks, such as *A. m. ligustica*, Buckfast, or *A. m. carnica*. Local adaptation is often expressed in a higher life expectancy of the native or local versus non-local stock (B  chler et al., 2014). Thus, when starting a breeding program to improve honey yield, gentleness, and potential further traits in a local population, one may ask whether this might impair the local adaptation. It is expected that the adapted populations benefit from their overall vitality, which, in most cases, is strongly associated with resistance to the *Varroa* mite and its secondary virus infections. Local adaptation not

only comprises resistance to *Varroa*, but may also involve a blend of many traits such as the adaptation of brood rearing to the local climatic and floristic conditions, an expression of defensive or adaptive behaviors against locally present predators (Zakour & Bienefeld, 2014) and pests, pathogens, or harsh weather conditions. However, we feel that for practical purposes the *Varroa* resistance can be taken as a reliable indicator of this complex. Selection for traits like honey yield and gentleness with a simultaneous emphasis on *Varroa* resistance can be seen as a safe way to improve locally adapted populations. For *Varroa* resistance, a palette of *Varroa* related traits for breeding has been recognized and methods for evaluating the expression have been developed (Table 1).

The complexities of the given traits reflect into the costs related to their evaluation. Recently, as part of the EurBeST project (www.eurbest.eu), the costs for colony evaluation in different European countries

were estimated. The costs for evaluation of varroa-related traits proved to make up the largest part of the overall costs for colony evaluation (Figure 1). Such findings will certainly play a role in breeders' decision-making, which traits to include in their programs towards the achievement of the breeding goal.

Finally, one should keep in mind that different breeding goals will, depending on the traits involved, have different prospects of success. The performance of a colony is determined in part by the bees' genetics and in part by other, external factors, and their interaction. The more heritable a trait is, i. e. the more it is influenced by genetics relative to environment, the easier it will be to achieve fast improvement. In contrast, traits with low heritability will take more generations to generate significant genetic progress. Furthermore, different traits also influence each other. For example, in some populations there appears to be a tendency that colonies that produce a lot of honey also

swarm easily, quantified by the genetic correlation between both traits (e.g., Brascamp et al., 2016). The traits "honey yield" and "low swarming tendency" are then in genetic conflict with each other. A comparable, or even more pronounced example, is the beekeepers' interest in selection of extensive spring brood rearing, which in turn has the undesired effect to facilitate the development of the *Varroa* population. Such a conflict does not mean that the two traits cannot be improved at the same time, but progress will be slowed down considerably. On the other hand, various traits can also reinforce each other, thus simplifying their common selection. Unfortunately, heritability and the ways in which traits influence each other (genetic correlations), generally differ between populations (e.g., Brascamp et al., 2016; Hoppe et al., 2020) and are usually not known before the initiation of a breeding program. One can start with a set of consensus parameters, but it is advisable to estimate genetic parameters after several generations of selection and make adjustments in the prioritisation of traits accordingly. Note that consensus parameters are amply available in farm animals with estimates in numerous populations and time frames, but that in honey bees, there are only very few reliable estimates available, hence consensus probably will be a merger of published values and common sense.

Table 1 A list of some *Varroa* related traits ranked by estimated complexity for evaluation.

Trait/parameter	Complexity of evaluation	Note
Natural mite fall	Low	Commonly used in many honey bee breeding programs
Adult bees' infestation	Low	
Hygienic behavior	Medium to high	Limited use in breeding programs
Grooming behavior	Medium to high	
REC - recapping	Medium to high	
Post-capping stage duration	High	
VSH – varroa sensitive hygiene	High	Limited use due to the dependence on laboratory/institute support
SMR – suppressed mite reproduction	High	

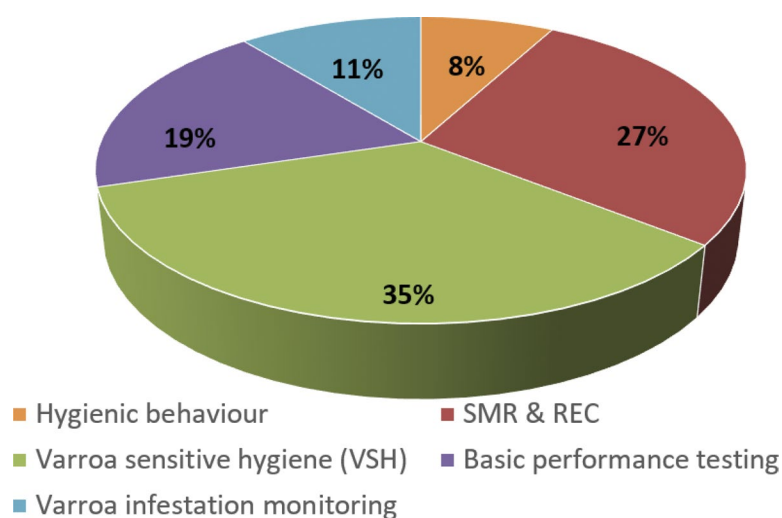


Figure 1. Share of different testing methods in colony evaluation costs in five EU countries, Germany, Italy, France, Greece and Poland (Büchler et al., 2022). SMR – Suppressed mite reproduction, REC – recapping. Basic performance testing stands for traditional traits/parameters such as honey yield, gentleness, calmness, swarming tendency etc.

The Initiation, Synchronisation, and Time-Line

Once the population of interest is identified and the breeding goal determined, the next question for future breeders is to address "When and how to start?". For this, there are different models, and none fits all scenarios. In fact, in most cases, a tailor-made model is chosen to meet the breeders' capacities, resources, ambitions and environmental conditions.

Everything starts with setting up test apiaries. Groups of sister queens should be distributed across the test apiaries as evenly as possible (Figure 2). The purpose of the distribution of sister queens across test locations is to facilitate the quantification of genetic and environmental differences between colonies. If colonies of sister queens are tested at one single apiary, it is more difficult – though possible with lower accuracy using pedigree information – to estimate what part of differences between colonies at different

apiaries is genetic, as environment and genetics are confounded. It is important that colonies within test apiaries are treated as equally as possible, because such harmonised practice enables fair comparison of the quality of colonies considering differences caused by the beekeeping techniques applied, weather conditions, food sources etc. (Büchler et al., 2013; Ruttner, 1972; Uzunov et al., 2015). Based on the recorded data, breeders need estimated breeding values to describe the relative genetic superiority or inferiority of the queens to be selected (Brascamp & Bijma, 2019; Hoppe et al., 2020). It goes beyond the scope of this paper to go into their computation, but we will use the term “breeding values” also if comparisons are restricted to deviations from test-apiary means.

The next question that beekeepers raised is “How long does it take to complete a single breeding cycle?”. The expected period for implementing all elements of the breeding cycle in Central Europe, characterised by well-defined seasonality, is around 24 months (Figure 3). The rearing and mating of the queens and establishing new test colonies in Central Europe are commonly done around mid to late spring (May-June) and in Western Europe a month later. Thus, following the bees’ genetic turnover in the colonies, the performance testing starts at the end of the summer (August) or at the latest before wintering. This segment of the breeding cycle usually finishes with the final honey harvesting in the next season. After that, the collected data are submitted to estimate breeding values, an element

that should be executed sufficiently long before the beginning of queen rearing in the following year to allow well-informed selection. With this queen rearing, the next cycle starts.

The given model shows that new queens’ testing starts every two years, a common approach (mono-cycle model) for the starters in honey bee breeding. One can, however, also start a new cycle each year, resulting in a dual-cycle model. In Figure 4, these two alternatives are illustrated. In both cases, breeding starts by setting up test colonies that are subjected to performance tests and breeding value estimation in the following year. What ensues is the selection of colonies from which to rear the next generation of queens the year after that. In these models we assume that the queens in the first year (year 0) are open mated such that selection only concerns the maternal side. Due to lack of implemented mating control, for some breeding programs, this is the only attainable approach.

In case of open mating, the genetic quality of the paternal side represents the average of colonies in a large surrounding. Beside performance testing, mating therefore is a crucial aspect of honey bee breeding as controlled mating allows selection on the paternal side on top of selection on the maternal side. Breeding schemes will be considerably more successful if mating control is applied (Plate et al., 2019). For this purpose, drone-producing colonies (DPC) are separately reared, usually as sister groups with a common dam (the so-called 4a queen, from its position in a standard pedigree scheme). In this article, we discuss breeding strategies that arise from mating control on isolated mating stations. Other strategies of mating control, such as instrumental insemination, may give rise to slightly different breeding plans.

Ideally, the dams of the queens of the DPC should be selected for their good genetic qualities to ensure genetic improvement by controlled mating. This clearly adds additional time to the generation turnover on the paternal side. If a breeding program is newly set up in year 0, the first performance tests take place in year 1 and the first breeding values are available in year 2. If one decides based on these breeding values, from which queens one should breed DPC, the first drones for controlled mating are available in year 3. In this way, it would only be in years 4 (mono-cycle) or 3 (dual-cycle) that breeding queens

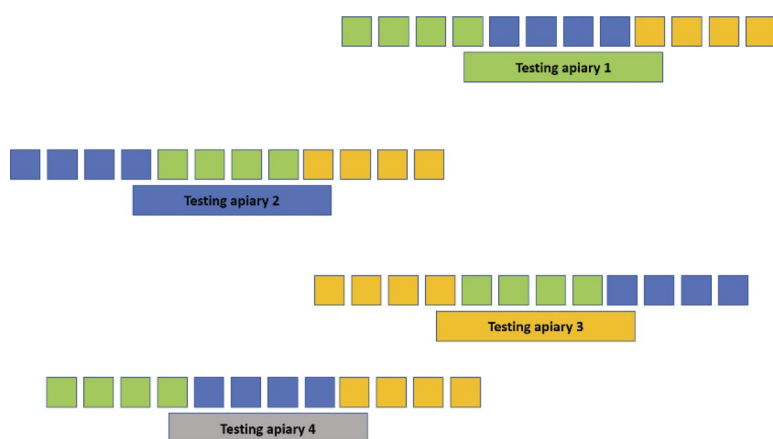


Figure 2. A simplified example of queens' exchange between test stations where the persons responsible for the testing apiaries 1, 2, and 3 are breeders, and apiary 4 is a tester. We recognise two types of beekeepers involved in the process, one is called breeder, who rears and may test queens, and another called tester and who only is doing testing. The tester's different colour from the colours of the sister queens, shows that there is no local genotype tested but only received ones from other breeders. Other alternative organisational schemes are also feasible in practice.

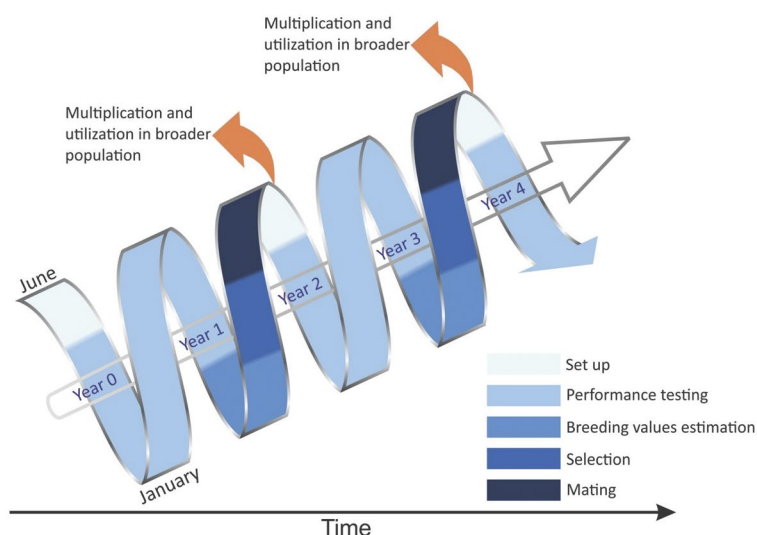


Figure 3. A 3D model of the breeding cycle spiral achieving selection progress in time.

can mate in a controlled manner. These options are depicted as scenarios C in Figure 4.

While these are the typical options for established breeding programs that have been running for a while, one may think of other options for the beginning of a new breeding program. In particular, it is interesting, how controlled mating of the new queens in year 2 (mono-cycle), or of years 1 and 2 (dual-cycle) can be achieved. To make this possible, one has to select 4a queens to produce the DPC either right after setting them up in year 0 (scenario A) or during the phase of performance tests in year 1 (scenario B).

When at initiation of a breeding program DPC are established in the same year in which new colonies are set up (scenario A), the breeder has only very limited information to select the dams of DPCs (the 4a queens). One can select queens that come from successful lines, based on the beekeepers' intuitions or notes, but no well-informed within-family selection of 4a queens is possible. If the establishment of DPC is delayed by one year (scenario B), there is already some more information on the 4a queens. At the very least, the breeder knows that they were able to survive one winter. Also, some information can already be drawn from the ongoing performance test, and while the breeder does not yet have estimated

breeding values, they can already see if a colony is well above the average and choose it as 4a queen. Finally, when DPCs are established after the breeding value estimation (scenario C), one can make the most accurate decision.

The degree of selection progress in the population is affected by the approach taken, with the clear advantage of the dual-cycle breeding model, which enables the selection and rearing of a new generation of queens every year. The dual-cycle breeding model allows breeders to exploit the genotypes from parallel testing generations rationally and make the selection best. Note that the typical time between generations (generation interval) in this case for the maternal side is two years and for the paternal side is three years. In contrast, mono-cycle breeding has a paternal generation interval of four years. As the annual genetic improvement is inversely proportional to the average generation interval of the maternal and paternal side, it is always relevant to consider whether shortening of generation intervals is possible. Such opportunities may exist for some honey bee populations from South and Mediterranean Europe, like accelerating the selection progress by making the best of the prolonged periods of brood and drone rearing. Let us take the extreme example of Malta and its endemic population, *A. m. ruttneri*. There are two main brood-rearing periods: first

from late February to mid-April, as the most prolific brood-rearing and swarming season and second from September to early November, which generally represents a season when some colonies mainly supersede queens (Uzunov et al., unpublished). Under such conditions, the implementing period for an entire breeding cycle could be possibly shortened due to the queen and drone rearing possibilities. This proves to be the main limiting factor for shortening the breeding cycle in mainland Europe because of the restricted single or short period. Such an opportunity might be possible at least for the traits such as gentleness, calmness and hygienic performance, but for honey production, swarming tendency and particularly *Varroa* related traits, it seems quite challenging. However, the aforementioned breeding group from Malta considers shifting from the present simple to a dual-cycle breeding model. Still, there are also difficulties, such as drought and hot summer, lack of food source, etc., which need to be considered and scrutinised before a decision is made. Nevertheless, a shorter breeding cycle might have its limitations coming from the insufficient time for data collection and, consequently, information for reliable estimation of breeding values. So, in the example given for Malta, it is unclear if shortening the generation interval is actually beneficial for annual genetic improvement for all traits. Such alternative strategies should

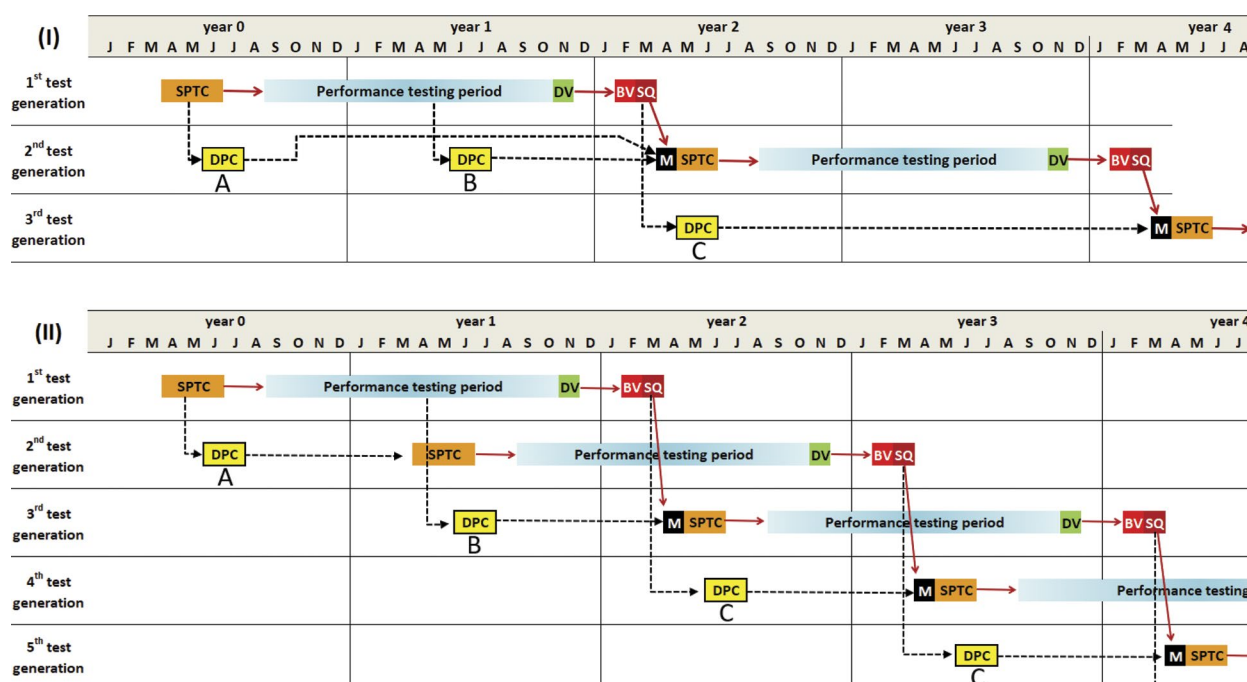


Figure 4. Models of (I) mono-cycle and (II) dual-cycle breeding when initiated under Central European conditions. SPTC – setting up performance testing colonies, DPC – drone-producing colonies, DV – Data collection and validation, BV – Breeding values estimation, SQ – selection and queen rearing, and M – Mating. The red solid arrow line represents the dams' route and the black dotted line represents the sires' route. A, B and C – scenarios of timing for establishing drone-producing colonies. Note that scenario C repeats as routine approach after year 2.

be verified by running simulation-based models as described by Plate et al. (2020).

As a last component, inbreeding has to be properly addressed. Inbreeding is a phenomenon emerging from the mating of (close) relatives, and it generally leads to so-called inbreeding depression, a deterioration of performance. The inbreeding coefficient of individuals can be estimated from the pedigree and combining these the rate of increase of inbreeding over time can be estimated as well. In a closed population, the degree of inbreeding always increases because potential parents gradually become more related. Nevertheless, there is consensus that an increase of inbreeding by 0.5 to 1% per generation is acceptable. In the scenarios described above, scenario C in the dual-cycle breeding scheme involves a lower risk of inbreeding than the other scenarios as the queens and DPC originate from different cycles. In general, one needs a sufficient number of parents to keep the increase in inbreeding sufficiently low. In the case of open mating, a rule of thumb might be that each year 50 dam queens are needed to raise the next generation of queens from which later on again 50 are selected. In the case of controlled mating the same minimum number of selected queens applies. However, here a more relevant parameter is the number of queens from which groups of DPCs are raised. This number should be at least around 20 (Plate et al., 2020).

The Chosen Ones

Once the data are collected and the breeding values for the traits and genotypes estimated, the breeders face the ultimate question, “Which ones to choose as the best parents for the new generation of queens?”. It seems that the answer is easy and straightforward since all tested queens are now listed and ranked based on their breeding values. However, in reality, this list is only the beginning and selection includes other aspects such as availability and relatedness. The top-ranked queens are of central interest for every breeder and beekeeper, but they may be no longer available to rear their daughters because in the meantime they got lost. If the elite queens are available during the rearing period, the decision is straightforward, but often some top queens, due to a variety of reasons, are lost after the performance testing or during the winter or early spring. That is why breeders, once the data

collection is done, should take care and precaution to ensure adequate conditions for the survival of the queens and colonies, which were meticulously and expensively tested for a period of more than a year. Still, if the best-ranked queen is lost, the breeders can select from the available higher-ranked queens.

As described earlier, the development of inbreeding in a population depends mainly on the numbers of selected queens and DPCs. However, to a lesser extent, it can also be influenced by the choice which particular colonies are selected. For example, by restricting the number of selected queens from the same sister group it can be avoided that dam queens are too closely related, thus lowering inbreeding rates in later generations. A general solution combining maximal response to selection with limited increase in inbreeding was developed by Meuwissen (1997). It results in more spread of parents than otherwise would be the case. Especially in mono-cyclic breeding schemes, some of the selected queens and DPCs will be closely related and one should avoid pairing them with one another because this would result in high inbreeding rates in the next generation. Even when the mating partners have been selected, a decision should be made about their mating roles, i.e., will they be used as a genetic source for queen rearing, drone-producing colonies, or both? In the widely used Beebreed breeding platform (www.beebreed.eu), an interactive online tool is available, where breeders and beekeepers can “play” with a selection of different mating partners and examine the expected breeding values and inbreeding coefficients of the potential offspring.

Besides the rearing of the individuals for the next testing cycle in the breeding program, an additional aspect is the opportunity to propagate the achieved genetic progress into the broader local honey bee population. In both the breeding program and in the propagation of genetic progress, mating control plays a decisive role. The intention is to address this in a forthcoming article as a part of the series of texts concerning the honey bee breeding programs.

Acknowledgement

We would like to thank Prof. Dr. Sreten Andonov, Dr. Andreas Hoppe, Mr. Thomas Galea, Dr. Marin Kovačić, Mr. Egoitz Galarza and Prof. Dr. Kaspar Bienefeld for their valuable comments and

suggestions for improvement of the readability of the article. We also thank Mr. Dejan Madjirov for the graphical design of Figure 3.

Disclosure statement

No potential conflict of interest was reported by the authors.

References

- Armbruster, L. (1919). *Bienenzüchtungskunde*. Th. Fisher, Leipzig, Germany.
- Bienefeld, K., Reinhardt, F., & Pirchner, F. (1989). Inbreeding effects of queen and workers on colony traits in the honey bee. *Apidologie*, 20(5), 439–450. <https://doi.org/10.1051/apido:19890509>
- Brascamp, E. V., & Bijma, P. (2019). Software to facilitate estimation of genetic parameters and breeding values for honey bees. *Proceedings of the 46th Apimondia*, 8–12 September 2019 in Montréal, Canada. <https://edepot.wur.nl/546515>.
- Brascamp, E. V., Willam, A., Boigenzahn, C., Bijma, P., & Veerkamp, R. F. (2016). Heritabilities and genetic correlations for honey yield, gentleness, calmness and swarming behaviour in Austrian honey bees. *Apidologie*, 47(6), 739–748. <https://doi.org/10.1007/s13592-016-0427-9>
- Büchler, R., Andonov, S., Bienefeld, K., Costa, C., Hatjina, F., Kezic, N., ... Wilde, J. (2013). Queen rearing and selection. In V. Diemann, J. D. Ellis, & P. Neumann (Eds.), *The COLLOSS BEEBOOK: Standard methods for Apis mellifera research*. *Journal of Apicultural Research*, 52, 1–30. <https://doi.org/10.3896/IBRA.1.52.1.07>
- Büchler, R., Costa, C., Hatjina, F., Andonov, S., Meixner, M. D., Conte, Y. L., Uzunov, A., Berg, S., Bienkowska, M., Bouga, M., Drazic, M., Dyrba, W., Kryger, P., Panasiuk, B., Pechhacker, H., Petrov, P., Kezic, N., Korpela, S., & Wilde, J. (2014). The influence of genetic origin and its interaction with environmental effects on the survival of *Apis mellifera* L. colonies in Europe. *Journal of Apicultural Research*, 53(2), 205–214. <https://doi.org/10.3896/IBRA.1.53.2.03>
- Büchler, R., Uzunov, A., Costa, C., Meixner, M., Le Conte, Y., Mondet, F., Kovacic, M., Andonov, S., Carreck, N. L., Dimitrov, L., Basso, B., Bienkowska, M., Dall'Olio, R., Hatjina, F., & Wirtz, U. (2022). EurBeST — A pilot study testing varroa-resistant bees under commercial beekeeping conditions. *American Bee Journal*, 162(2), 213. <https://doi.org/10.2762/470707>
- Hoppe, A., Du, M., Bernstein, R., Tiesler, F. K., Kärcher, M., & Bienefeld, K. (2020). Substantial genetic progress in the international *Apis mellifera carnica* population since the implementation of genetic evaluation. *Insects*, 11(11), 768. <https://doi.org/10.3390/insects11110768>
- Kovačić, M., Puškadija, Z., Dražić, M. M., Uzunov, A., Meixner, M. D., & Büchler, R. (2020). Effects of selection and local adaptation on resilience and economic suitability in *Apis mellifera carnica*. *Apidologie*, 51(6), 1062–1073. <https://doi.org/10.1007/s13592-020-00783-0>
- Meixner, M. D., Büchler, R., Costa, C., Andonov, S., Bienkowska, M., & Bouga, M. (2015). Looking for the best bee. An experiment about interactions between origin and environment of honey bee strains in Europe. *American Bee Journal*, 155, 663–666.
- Meuwissen, T. H. E. (1997). Maximizing the response of selection with a predefined rate of inbreeding. *Journal of animal science*, 75(4), 934–940. <https://doi.org/10.2527/1997.754934x>

Momeni, J., Parejo, M., Nielsen, R. O., Langa, J., Montes, I., Papoutsis, L., Farajzadeh, L., Bendixen, C., Căuia, E., Charrière, J.-D., Coffey, M. F., Costa, C., Dall'Olio, R., De la Rúa, P., Drazic, M. M., Filipi, J., Galea, T., Golubovski, M., Gregorc, A., ... Estonba, A. (2021). Authoritative subspecies diagnosis tool for European honey bees based on ancestry informative SNPs. *BMC Genom*, 22(1), 101. <https://doi.org/10.1186/s12864-021-07379-7>.

Plate, M., Bernstein, R., Hoppe, A., & Bienefeld, K. (2019). The importance of controlled mating in honeybee breeding. *Genetics, Selection, Evolution: GSE*, 51(1), 74. <https://doi.org/10.1186/s12711-019-0518-y>

Plate, M., Bernstein, R., Hoppe, A., & Bienefeld, K. (2020). Long-term evaluation of breeding scheme alternatives for endangered honeybee subspecies. *Insects*, 11(7), 404. <https://doi.org/10.3390/insects11070404>

Ruttner, H. (1972). Technical recommendations for methods of evaluating performance of bee colonies. In F. Ruttner (Ed.), *Controlled mating and selection of the honey bee* (pp. 87–92). Apimondia Publishing House.

Uzunov, A., Brascamp, E. W., & Büchler, R. (2017). The basic concept of honey bee breeding programs. *Bee World*, 94(3), 84–87. <https://doi.org/10.1080/0005772X.2017.1345427>

Uzunov, A., Büchler, R., & Bienefeld, K. (2015). *Performance testing protocol. A guide for european honey bee breeders* (1st ed.). Kirchhain, Germany: Landesbetrieb Landwirtschaft. <https://doi.org/10.13140/RG.2.2.1829.86244>

Zakour, M. K., & Bienefeld, K. (2014). Basic considerations in the development of breeding plans for honey bees, illustrated by data on the native Syrian honey bee (*Apis mellifera syriaca*). *Journal of Apicultural Research*, 53(2), 314–326. <https://doi.org/10.3896/IBRA.1.53.2.13>

Aleksandar Uzunov*

Faculty of Agricultural Sciences and Food,
University Ss. Cyril and Methodius in
Skopje, Skopje, Macedonia

*Email: uzunov@fzh.ukim.edu.mk

<http://orcid.org/0000-0003-1240-868X>

Evert W. Brascamp

Wageningen University & Research
Animal Breeding and Genomics,
Wageningen, The Netherlands

<http://orcid.org/0000-0003-2284-0513>

Manuel Du

Institute for Bee Research Hohen
Neuendorf, Hohen Neuendorf, Germany
<http://orcid.org/0000-0003-2264-1610>

Ralph Büchler

Landesbetrieb Landwirtschaft Hessen -
Bieneninstitut, Kirchhain, Germany
<http://orcid.org/0000-0001-7749-1870>

Introducing the first English translation of:

'Keeping Bees simply and Respectfully'

By Johannes Wirz and Norbert Poeplau.

Translated from the German 2020 edition by Martin Kunz.

Jointly published by IBRA & Northern Bee Books,
2021. Paperback, 170 pages.

ISBN: 978-1-913811-03-7.

This comprehensive guide to beekeeping with the Golden Hive ensures that beekeeping is a joyful endeavour. At the same time our respect for the bees will continue to grow.

Dr Johannes Wirz, a board member of Mellifera, and master beekeeper Norbert Poeplau provide an introduction to the concept of bee friendly beekeeping and insights into the fascinating life of the makers of honey. The authors' experience guides the reader step by step while managing a Golden Hive throughout the course of a year.



AVAILABLE TO BUY FROM IBRA.ORG.UK/SHOP